DIRECT RESPONSE TO ARC REQUEST FOR BUILDERS · FRONTIER 1

Programmable Commercial Agreements on Arc L1

Turning multi-party commercial contracts into self-executing software with native USDC gas, sub-second finality, and mutual anti-ghosting guarantees.

0.50%	~350ms	$0.0001	100%
PROTOCOL FEE (VS 20% WEB2)	BFT SETTLEMENT FINALITY	NATIVE USDC GAS COST	NON-CUSTODIAL TRUST

The $2.4 Trillion Commercial Trust Dilemma

Digital labor and machine services move at internet speed, but the financial layer holding them together is predatory, slow, and fractured.

Web2 Freelance Rakes

Platforms like Upwork and Fiverr extract **10% to 20% rakes**, freeze contractor earnings in mandatory 5-to-14 day clearing holds, and resolve disputes through opaque, centralized customer service desks that default against independent workers.

`20% Platform Cut` `14–Day Clearing Hold`

Direct Crypto Risk

Sending raw crypto transfers forces **100% counterparty risk** onto one party. Either the buyer pays upfront and risks contractor abandonment, or the worker delivers upfront and risks non-payment. There is no middle ground.

`Zero Escrow Guarantee` `Counterparty Ghosting`

Legacy Chain Gas Floor

Ethereum L1 and legacy rollups require users to hold volatile native tokens (ETH) and incur $1 to $15 transaction fees. This makes micro-milestones ($10–$50 tasks) and machine-to-machine micropayments completely unviable.

`Volatile Gas Token` `Micropayments Impossible`

Troth: Programmable Commercial Escrow on Arc

An immutable, non-custodial smart contract infrastructure that converts commercial agreements into deterministic software logic over native USDC rails.

STEP 01

Agreement Lockup

Buyer deposits total USDC into non-custodial contract tranches (Milestone 1, 2, 3). Funds are cryptographically secured.

STEP 02

Claimable Links

Recipients bind via direct Arc address OR zero-friction claimable invite link with cryptographic hash authorization.

STEP 03

Dual-Timer Protection

Contractor delivers work. Client has 7-day review window. If client ghosts, funds auto-release; if contractor misses deadline, buyer can refund.

STEP 04

Sub-Second Settlement

Upon client approval or timer expiration, USDC disbursements execute in ~350ms with sub-cent gas fees.

Dual Recipient Flexibility

Supports instant wallet-to-wallet contracts or Web2-friendly claim links where contractors don't even need a wallet prior to starting work.

Public Zero-Wallet Verification Rail

Anyone can inspect an agreement status, milestone deliverables, and transaction receipts on our public audit rail with zero wallet connection.

Why Arc L1 is Troth's Fundamental Moat

Troth cannot exist on Ethereum or general-purpose rollups. Arc's native architectural primitives unlock real commercial viability.

> *"Turn commercial agreements into software: milestone releases, dispute workflows, and conditional settlement written as contract logic over USDC payment rails."*
>
> **— Arc Official Request for Builders, Frontier 1: Programmable Trade Workflows**

Native USDC Gas

Users pay gas directly in digital dollars (~$0.0001 USDC). Eliminates the dual-token friction where businesses must buy and hold ETH or SOL just to execute an escrow contract.

Sub-Second Deterministic Finality

Arc's Malachite BFT consensus delivers deterministic ~350ms finality. Milestone completions feel instantaneous, mirroring real-time card settlement without chargeback risk.

Agentic Machine Commerce

Directly ready for Circle Agent Stack. Autonomous AI agents can negotiate deliverable specs, lock budgets into Troth, verify task completion, and settle payments programmatically.

Live Code, Verified Contracts, Zero Vaporware

Troth is not a pitch deck concept. It is a live, deployed, and rigorously tested production application on Arc L1.

Technical Ground Truth

```
// Arc Layer 1 Verified Contract Address 0xF7A20db6E51171263856f282ed188f795206d649
// Malachite BFT Finality ~350ms deterministic state transition // Foundry Security
Test Suite 11 / 11 Unit Tests Passing (100% Coverage) ✓ testCreateAgreement() ✓
testClaimAgreement() ✓ testSubmitAndReleaseMilestone() ✓
testReviewTimeoutAutoRelease() ✓ testContractorGhostingRefund()
```

Shipped Surfaces

🌐 **Live Production Web Application:**
https://trytroth.netlify.app

🔍 **Public Zero-Wallet Verification Rail:**
https://trytroth.netlify.app/agreement/104

🎥 **100% Feature Coverage Video Walkthrough:**
https://trytroth.netlify.app/demo-video.mp4

📦 **Clean Open-Source Foundry Repository:**
https://github.com/A-Raphie/troth

Business Model & Economic Flywheel

By undercutting legacy rent-extracting platforms by 80% while offering superior speed and security, Troth captures high-velocity commercial transaction volume.

0.50%

Take Rate on Settlement

Deducted automatically upon successful milestone disbursement. Leaves 99.5% in contractor pockets while generating sustainable protocol treasury yield.

$100M+

Target Transaction Volume

At $100M annualized GMV, Troth generates **$500,000 ARR** in pure protocol revenue with near-zero marginal cloud infrastructure overhead.

Machine

Agentic Micropayments

High-frequency $5–$50 autonomous AI agent service settlements create compounding network velocity unachievable in legacy banking rails.

Target User Acquisition Channels

1. Cross-Border Dev Agencies: Frictionless international contractor settlements without SWIFT wire fees or FX spreads.

2. Web3 DAO Grant Programs: Milestone-gated contributor disbursements with automated review timers.

3. AI Agent Marketplaces: Programmatic escrow integration via REST & TypeScript SDKs using Circle Agent Stack.

Technical & Ecosystem Roadmap

Clear execution milestones designed to take Troth from live testnet prototype to the core settlement layer for trade on Arc.

MILESTONE 01 · Q4 2026

Arc Mainnet & Multichain

- Deploy verified contracts on Arc Mainnet (Chain ID 5042)
- Integrate **Circle Gateway** for frictionless cross-chain USDC deposits from Arbitrum, Base, and Ethereum
- Formal smart contract security audit with top-tier firm

`Circle Gateway` `Mainnet Launch`

MILESTONE 02 · Q1 2027

Agentic Trade API

- Full integration with **Circle Agent Stack** for autonomous AI contractor settlements
- REST / TypeScript SDK for B2B platforms and AI agents
- Multi-signature decentralized arbitration council module for disputed milestones

`Circle Agent Stack` `B2B Trade SDK`

MILESTONE 03 · Q2 2027

Institutional Trade Credit

- Invoice factoring & escrow yield streaming via **Circle USYC** tokenized yield
- Enterprise dashboard for contractor payroll and tax compliance
- Target $10M+ monthly settlement volume

`USYC Yield` `Invoice Factoring`

Partnership & Capital Request

We are seeking strategic backing from Circle and Arc to establish Troth as the default commercial agreement protocol on Arc L1.

CIRCLE DEVELOPER GRANTS

$50,000 USDC

Milestone-Based Grant Allocation:

- **$25,000 USDC:** Arc Mainnet deployment, Circle Gateway cross-chain deposit integration, and third-party security audit.
- **$25,000 USDC:** Circle Agent Stack A2A trade settlement integration, enterprise SDK release, and onboarding the first 100 commercial contractor teams.

ARC BUILDERS FUND & CIRCLE VENTURES

Seed Round (Up to $1.0M)

Strategic Equity / Token Investment:

- Dedicated core engineering support and co-marketing with Arc and Circle ecosystem teams.
- Direct distribution through Arc House and integration into Arc Portal as a flagship financial primitive.
- Scaling enterprise B2B sales and global compliance infrastructure.

Lead Contact & Founder: Raphael (Raphie)

GitHub: github.com/A-Raphie · Project Repo: github.com/A-Raphie/troth

trytroth.netlify.app